

# **Fairness, Normalization & Bias Reduction**

## **Day 15 Internship Deliverables**

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Submitted To

Company: Zecpath

Department: Product & Engineering Team

# 1.Fair scoring improvements

```
fairness_engine > fairness.py > ...
1  from normalize import normalize_resume
2  from bias_detector import detect_bias
3
4  sample_resume = {
5      "name": "Shinas PN",
6      "gender": "Male",
7      "age": 22,
8      "email": "SHINASP@GMAIL.COM ",
9      "phone": "9876543210 ",
10     "skills": [
11         "Python",
12         "python",
13         "Machine Learning",
14         "AI",
15         "ai"
16     ],
17     "education": "bca artificial intelligence",
18     "experience": 2
19 }
20
21 print("===== FAIRNESS REPORT =====\n")
22
23 # Normalize the resume
24 normalized = normalize_resume(sample_resume)
25
26 print("Normalized Resume")
```

```
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> cd fairness_engine
>> python fairness.py
===== FAIRNESS REPORT =====

Normalized Resume
{'name': 'Shinas Pn', 'email': 'shinaspn@gmail.com', 'phone': '9876543210', 'skills': ['ai', 'machine learning', 'python'], 'education': 'Bca Artificial
ntelligence', 'experience': 2}

-----
Sensitive Fields Detected:
- name
- gender
- age

-----
Fairness Status: PASS
Resume standardized successfully.
Bias indicators identified for masking.
```

## 2.Normalization logic

```
fairness_engine > normalize.py > ...
1  import json
2
3  def normalize_resume(resume):
4      """
5      Standardizes resume fields into a consistent format.
6      """
7
8      normalized = {
9          "name": resume.get("name", "").strip().title(),
10         "email": resume.get("email", "").strip().lower(),
11         "phone": resume.get("phone", "").strip(),
12         "skills": sorted(
13             list(set(skill.strip().lower() for skill in resume.get("skills", [])))
14         ),
15         "education": resume.get("education", "").strip().title(),
16         "experience": max(0, resume.get("experience", 0))
17     }
18
19     return normalized
20
21
22 if __name__ == "__main__":
23
24     sample_resume = {
25         "name": "shinas pn",
26         "email": "SHINASP@GMAIL.COM ",
27         "phone": "9876543210 ",
28         "skills": [
29             "Python",
30             "python"
```

### 3. Bias-reduction documentation

```
fairness_engine > bias_detector.py > detect_bias
1  import json
2
3  # Fields that should not influence AI hiring decisions
4  SENSITIVE_FIELDS = [
5      "name",
6      "gender",
7      "age",
8      "date_of_birth",
9      "religion",
10     "nationality",
11     "marital_status",
12     "photo"
13 ]
14
15 def detect_bias(resume):
16     detected = []
17
18     for field in SENSITIVE_FIELDS:
19         if field in resume:
20             detected.append(field)
21
22     return detected
23
24
25 if __name__ == "__main__":
26
27     sample_resume = {
28         "name": "Shinas PN",
29         "age": 22,
30         "gender": "Male",
```

